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Coimbatore, Tamil Nadu, India

EDUCATION

B.E. in Artificial Intelligence & Machine Learning

2025 – Present

KPR Institute of Engineering and Technology (KPRIET) | Coimbatore, India

Higher Secondary Education (Class XII)

Carmel Garden Matriculation Higher Secondary School | Coimbatore, India

EXPERIENCE

Machine Learning Intern

Jun 2026 – Jul 2026

Code Alpha | Coimbatore, Tamil Nadu (Remote)

- ▶ Built and deployed three ML dashboards (Fraud Detection, Disease Prediction, Credit Scoring); awarded Certificate of Completion (ID: CA/DF1/108253).

Data Science Intern

Jun 2026 – Jul 2026

Thiranex | Coimbatore, Tamil Nadu (Remote)

- ▶ Worked on applied data science tasks spanning data analysis, model building, and AI-driven feature development.

Machine Learning Intern

Apr 2026

Unlox & Tailyo Technologies (Job Bridge Initiative) | Remote

- ▶ Completed project-based ML internship on industrial workflows and real-world deployment; awarded Unlox Edge Certification (#UNXML-INT-9906).

PROJECTS

ML-Powered Service Recommendation System [\[Code\]](#)

Python, Scikit-learn, Pandas, NumPy

- ▶ Engineered a collaborative filtering engine using Cosine Similarity and Matrix Factorization, reducing RMSE scores on validation sets.
- ▶ Architected backend pipelines for data ingestion and model inference, optimizing transformation stage data flow.

Real-Time Object Detection YOLO Architecture [\[Code\]](#)

YOLOv8/v10, PyTorch, OpenCV, Roboflow

- ▶ Engineered an end-to-end CV pipeline achieving competitive Mean Average Precision (mAP) on domain-specific object classes.
- ▶ Optimized inference throughput by tuning hyperparameters, balancing mAP accuracy against FPS for edge-device deployment.

Banking Transaction Analysis & Fraud Pattern Detection [\[Code\]](#)

Python, Streamlit, Pandas, Plotly

- ▶ Built an interactive Streamlit dashboard analyzing 284K+ credit card transactions, exposing live fraud patterns via KPI metrics and risk tools.

Medical Diagnostic AI Dashboard - Disease Prediction [\[Code\]](#)

Python, Streamlit, Scikit-learn, XGBoost

- ▶ Evaluated XGBoost, Random Forest, and Logistic Regression pipelines via 5-Fold Cross-Validation, prioritizing Recall to minimize false negatives.

Nexus Bank AI Credit Scoring Engine [\[Code\]](#)

Python, Streamlit, Scikit-learn, Random Forest

- ▶ Designed a real-time underwriting dashboard with Debt-to-Income ratios and a FICO-style credit score gauge, trained on a synthetic dataset.

DIY Encrypted Messenger (CircuitMess Chatter 2.0)

C++, Arduino IDE, LoRa, ESP32

- ▶ Architected a private off-grid communication device on ESP32 with LoRa, achieving long-range encrypted data transmission.
- ▶ Engineered end-to-end AES encryption protocols within C++ firmware to ensure confidentiality across hardware channels.

TECHNICAL SKILLS

Languages: Python, C++, SQL

AI/ML & CV: PyTorch, YOLO (v8/v10), Scikit-learn, XGBoost, OpenCV, Pandas, NumPy

Data & Visualization: Streamlit, Plotly, GridSearchCV, Joblib

Tools & Hardware: Git, Linux, Arduino IDE, LoRa, ESP32, Roboflow

CERTIFICATIONS

Machine Learning Course Completion — Unlox Edge (#UNXML-2025-9906)

Apr 2026

Machine Learning Virtual Internship — CodeAlpha (ID: CA/DF1/108253)

Jun 2026

Introduction to Programming & Python — Kaggle

Nov 2025

C Programming Basics — Simplilearn (ID: 9020864)

Sep 2025

CORE COMPETENCIES

Machine Learning • Computer Vision • Embedded Systems • Algorithm Optimization • Hardware-Software Integration • Data Preprocessing • Model Evaluation